Parasympathomimetic drugs (Cholinomimetics)

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Objectives

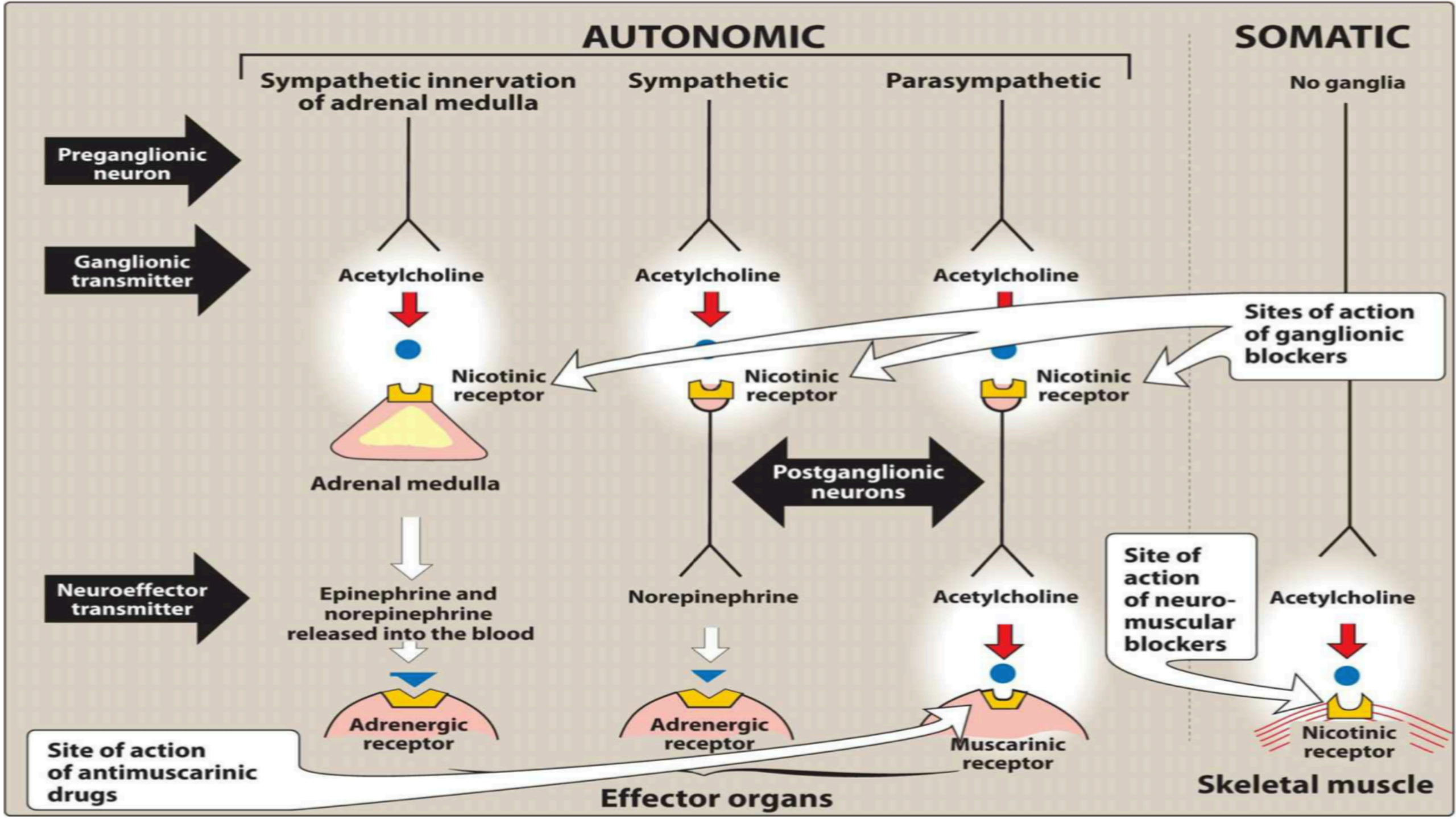
- Identify mechanism of action of cholinomimetics
- Know the types of receptors
- List name of drugs acting on the receptors with their indications

- Drugs can affect cholinergic transmission either by acting on postsynaptic ACh receptors as agonists or antagonists or by affecting the release or destruction of endogenous ACh.

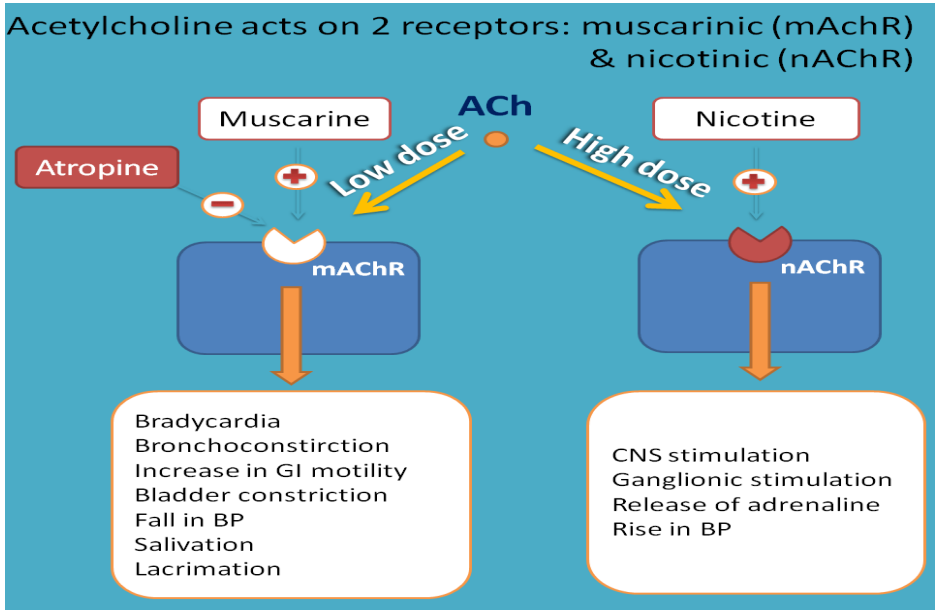
- The parent compound of all cholinomimetic drugs is **acetylcholine (Ach)**.

ACh is the natural neurotransmitter at the following sites:

- All autonomic ganglia whether sympathetic or parasympathetic.
- Parasympathetic nerve endings to involuntary organs and exocrine glands.
- Sympathetic nerve endings to thermoregulatory sweat glands.
- Sympathetic nerve endings to adrenal medulla.
- Skeletal muscle motor end plate.
- Certain tracts within the CNS


AUTONOMIC
SOMATIC
Sympathetic innervation of adrenal medulla
Sympathetic
Parasympathetic
No ganglia
Preganglionic neuron
Ganglionic transmitter
Acetylcholine
Acetylcholine
Acetylcholine
Sites of action of ganglionic blockers
Nicotinic receptor
Nicotinic receptor
Nicotinic receptor
Adrenal medulla
Postganglionic neurons
Neuroeffector transmitter
Epinephrine and norepinephrine released into the blood
Norepinephrine
Acetylcholine
Site of action of neuro-muscular blockers
Acetylcholine
Adrenergic receptor
Adrenergic receptor
Muscarinic receptor
Nicotinic receptor
Site of action of antimuscarinic drugs
Effector organs
Skeletal muscle

- ACh acts on both **muscarinic** and **nicotinic** receptors.
- ACh is **not used clinically** because: (1) it has very short duration of action (seconds) due to rapid hydrolysis by AChE (Acetylcholine esterase) enzyme; and, (2) it lacks selectivity.
- **Cholinomimetic drugs** are drugs that produce effects similar to ACh or cholinergic nerve stimulation, but with **more selectivity** and **fewer side effects** than ACh.



Mechanism of action

There are 2 distinct classes of receptors for Ach defined on the basis of their activation by alkaloids:

1.Muscarinic receptors (activated by muscarine) which are of 5 subtypes:

a.M1-in gastric parietal cells and CNS

b.M2-in heart, nerves, smooth muscles

c.M3-in exocrine glands, smooth muscles and bladder

d.M4 & M5-in CNS

2.Nicotinic receptors (activated by nicotine) - are found at neuromuscular junctions and their stimulation causes muscle fasciculation.

Classification of cholinomimetic drugs

Direct-acting cholinomimetics	Indirect-acting cholinomimetics
They act by **direct stimulation** of cholinergic receptors	They act by **inhibition of AChE enzyme** leading to accumulation of ACh.
■ **Muscarinic agonists** Bethanecol, carbachol Pilocarpine, cevimeline	■ **Reversible ChE inhibitors:** Physostigmine, neostigmine, pyridostigmine, donepezil
■ **Nicotinic agonists** Nicotine, lobeline	■ **Irreversible ChE inhibitors:** Organophosphate compounds

I-DIRECT-ACTING PARASYMPATHOMIMETICS

Muscarinic agonists:

Pharmacological effects:
CVS :

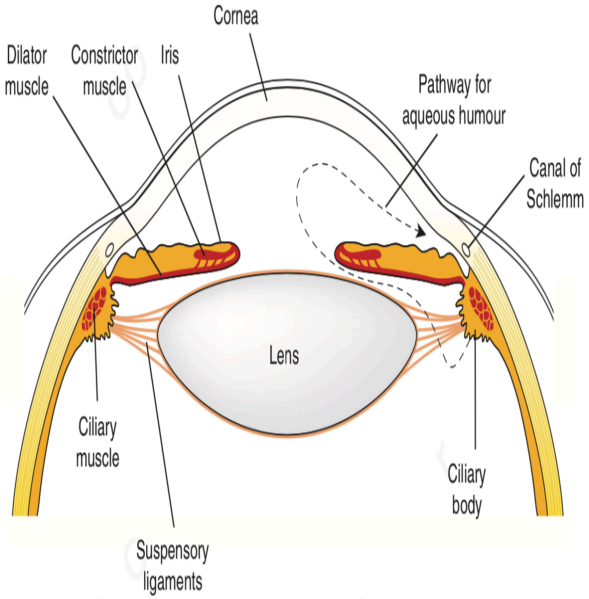
- Decreased heart rate (SA node) (M2).
- Vasodilation of all vascular beds through release of EDRF(endothelium derived relaxing factor) (M3).

Respiratory:

- Contraction of bronchial smooth muscle(M3).
- Increase bronchial secretions (M3).

Eye:

- **Miosis** due to contraction of constrictor pupillae muscle (M3).
- Accommodation for **near vision** due to contraction of ciliary muscle (M3).
- Decrease IOP (intraocular pressure) (contraction of the ciliary muscle causing opening of the trabecular meshwork and facilitates drainage of aqueous humor).



GIT:

- Increase motility and relaxation of sphincters (M3).
- Salivation (M3) and increase HCl secretion (M1).

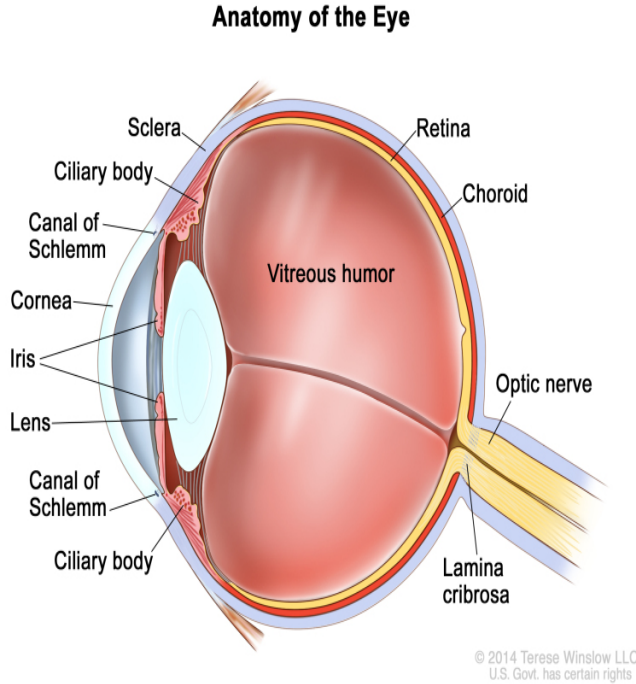
Urinary tract:

- Contraction of bladder smooth muscles (M3).
- Relaxation of sphincters (M3).

- **Exocrine glands:**
- Increase all exocrine secretions, salivation, lacrimation, sweating, etc.

1. Carbachol

- It is a choline ester but resistant to hydrolysis by Aceylcholine esterase (AChE) enzyme.
- It stimulates both **muscarinic** and **nicotinic** receptors
- Because of its high potency, receptor nonselectivity, and relatively long duration, carbachol is rarely used systemically.
- It is used as **eye drops** to ↓ IOP in **glaucoma**. It contracts the ciliary muscles causing opening of the trabecular meshwork and facilitates drainage of aqueous humor.



2. Bethanecol

- It is a choline ester but resistant to hydrolysis by AChE enzyme, so it has long duration of action (2-3 h) as compared to Ach.
- It stimulates **muscarinic** receptors with **no activity on nicotinic receptors**.
- It is used to reverse **post-operative urine retention**.
- May also be used to treat neurogenic atony as well as megacolon.

3. Pilocarpine

- It is a natural plant alkaloid.
- Acts as a **muscarinic agonist** with **no nicotinic effects.**
- Can be given orally to **increase salivary secretion** and decrease symptoms of dry mouth (xerostomia) associated with **Sjögren syndrome.**
- Pilocarpine is used as **eye drops** to decrease IOP in **glaucoma**.

4. Methacholine

- Used to assist in the diagnosis of asthma due to its bronchoconstricting properties.

Adverse effects of muscarinic agonists

- Most important side effects include nausea, vomiting, sweating, salivation, **bronchoconstriction**, hypotension, and diarrhea; all of which can be blocked by **atropine**.

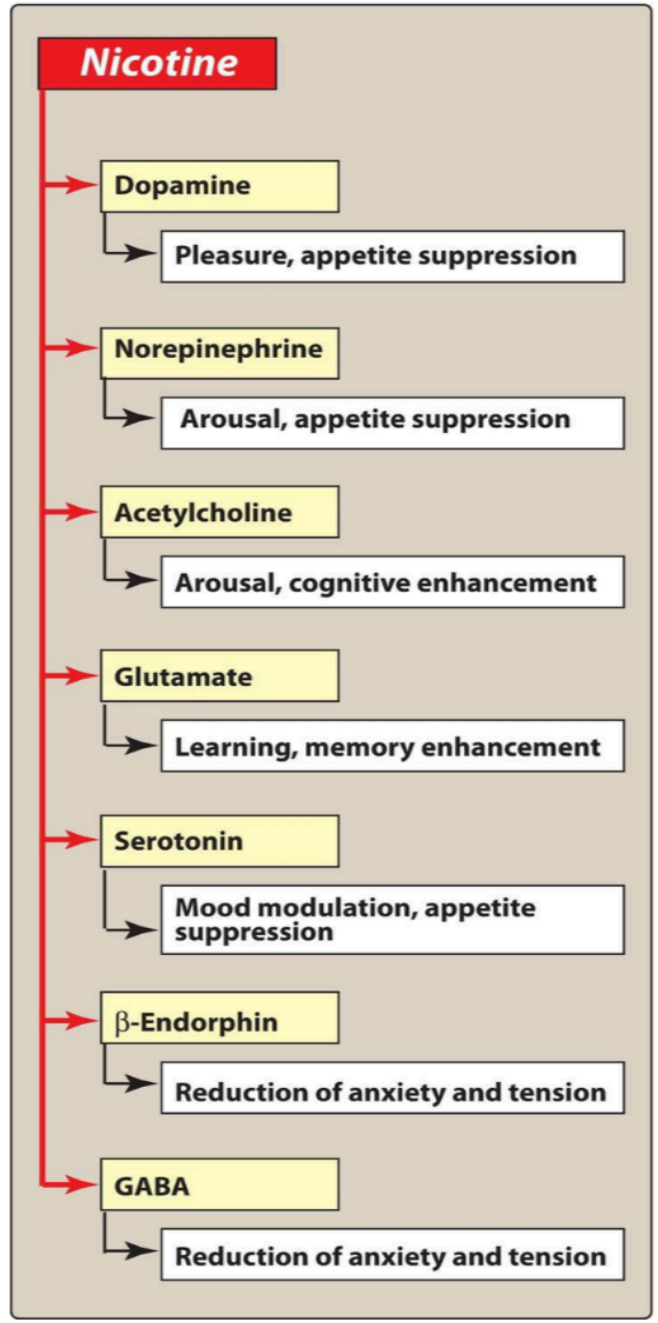
Contraindications of muscarinic agonists

- Peptic ulcer
- Bronchial asthma
- Heart block

Nicotinic agonists:

Nicotine

- It is a component of cigarette smoke. It is a **poison** with many adverse effects and **no therapeutic** benefit.
- The overall effects of nicotine are complex and result from **mixed** stimulation and inhibition of all autonomic ganglia:
- ***Small doses*** stimulate autonomic ganglia leading to hypertension, tachycardia, increase GIT peristalsis, increase HCl secretion, and CNS stimulation.
- ***Toxic doses*** lead to hypotension and CNS depression due to ganglion blockade.
- Nicotine is an addictive substance. **Transdermal patches** containing nicotine are used to help smokers stop smoking.



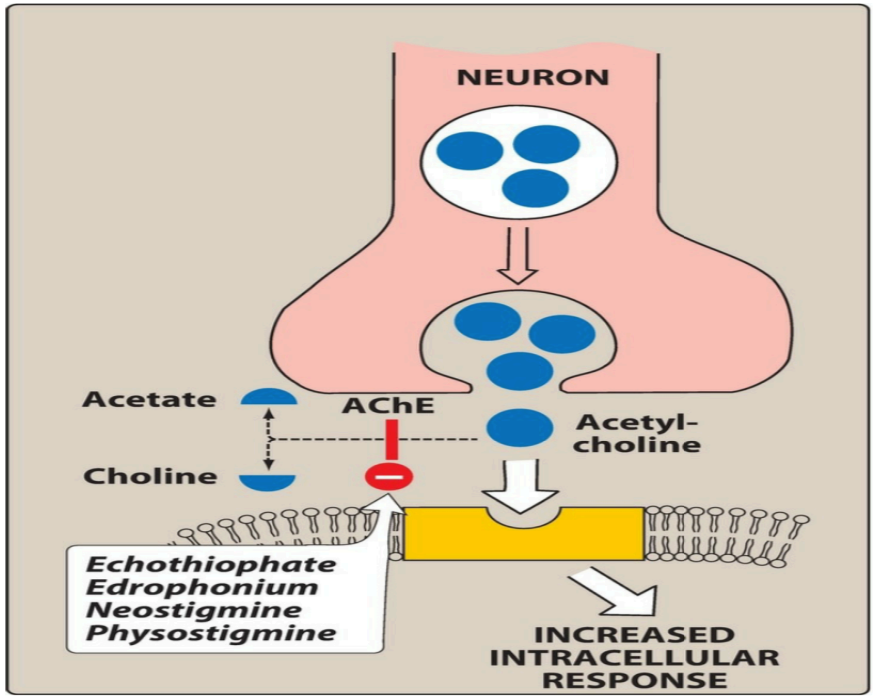
II-INDIRECT-ACTING PARASYMPATHOMIMETICS (Cholinesterase inhibitors)

Mechanism and pharmacological effects

- Indirect-acting parasympathomimetics **inhibit AChE enzyme** resulting in accumulation of ACh and stimulation of both **muscarinic** and **nicotinic** receptors.
- They are classified, according to nature and duration of **AChE inhibition,** into **reversible** and **irreversible** inhibitors.

II-a: Reversible AChE inhibitors:

- They interact with AChE enzyme by making a **reversible bond** allowing a duration of inhibition lasting from minutes to hours.



1. Physostigmine

Natural plant alkaloid (tertiary amine) that is well-absorbed from the GIT and **can pass to CNS**.

It can reversibly inhibit AChE enzyme for 3-4 hours, leading to:

- **Muscarinic effects*:*** hypotension, bradycardia, salivation, lacrimation, increased GIT peristalsis (diarrhea and colic), miosis, etc.
- **Nicotinic effects:** skeletal muscle contraction.
- **Central effects:** headache, insomnia, excitation, and convulsions.

Therapeutic uses

- Because of lack of selectivity and harmful CNS effects, it is usually used as **eye drops** to produce miosis and treat chronic **glaucoma**.
- Physostigmine can be used to reverse the central and peripheral manifestations of **atropine poisoning**.

2. Neostigmine

- Synthetic drug (quaternary amine) that is **poorly absorbed** from the GIT and **cannot pass to CNS**.
- It is similar to physostigmine in mechanism and effects but it has **no CNS actions**

Therapeutic uses

- To reverse postoperative **urine retention** and **paralytic ileus**. It is contraindicated if there is **mechanical obstruction** (to avoid rupture of the bladder or intestine).
- To reverse postoperative **muscle paralysis** resulting from the use of non- depolarizing neuromuscular blockers.
- Treatment of **myasthenia gravis:** Neostigmine not only increases ACh level in the neuromuscular junction but also can **directly stimulate** nicotinic receptors at the motor end plate.

3. Pyridostigmine

- Reversible AChE inhibitor similar to neostigmine.

It is more preferred than neostigmine in the chronic treatment of **myasthenia gravis** because:

- It has more selective action on neuromuscular junction (fewer unwanted muscarinic effects).
- It has longer duration of action than neostigmine.

4. Edrophonium

- It acts similar to neostigmine and pyridostigmine but has very short duration of action (5-15 minutes).
- It is used in the **diagnosis of myasthenia gravis.**

5. Donepezil, Rivastigmine and Galantamine

- They are AChE inhibitors that act more selectively on **central AChE** enzyme.
- Patients with Alzheimer's disease have a deficiency of cholinergic neurons and therefore lower levels of Ach in the CNS.
- They are used to increase ACh levels in the CNS and thus improve memory and cognitive deficit associated with **Alzheimer's disease**.

II-b: Irreversible AChE inhibitors:



- Organophosphate compounds
- They include:
  - **Drugs:** echothiophate eye drops
  - **Insecticides:** parathion and malathion
  - **Nerve gases:** tabun, sarin and soman



- Organophosphates are **highly lipid soluble** and rapidly absorbed by all routes including the **skin**.
- Their **CNS** penetration is rapid and high.
- They interact with AChE enzyme by making **irreversible (covalent) bond** (i.e. phosphorylation of the enzyme).
- Once AChE is inhibited, ACh accumulates throughout the nervous system, causing muscarinic and nicotinic symptoms.
- **Echothiophate** is the only non-absorbable organophosphate. It is available as miotic **eye drops** for **glaucoma**. Its effect in the eye lasts for weeks.

Manifestations of organophosphate toxicity:

- **CVS:** hypotension, bradycardia, sweating.
- **Respiratory:** bronchospasm, increase bronchial secretions, respiratory muscle paralysis.
- **GIT:** abdominal colic, diarrhea, and salivation.
- **Eye:** severe miosis *(pinpoint pupil)*, lacrimation.
- **CNS:** hallucinations, convulsions, and coma.
- **Skeletal muscle:** twitches and fasciculation.
- **The cause of death** is respiratory failure (blocked airway, paralyzed respiratory muscle & inhibited respiratory centre).

Management

- Ensure patent airway and artificial respiration.
- Gastric lavage and skin wash to remove the toxin.
- Intravenous normal saline to raise BP.
- **The triad: atropine – pralidoxime – diazepam**

References

- Mansura clinical pharmacology for medical students
- Lippincott Illustrated Review - Pharmacology 7th Edition (2018)-3

- Further reading: https://www.ncbi.nlm.nih.gov/books/NBK430685/?term=muscarinic%20agonist